

Supplement: A Motivation-to-Color Readout for the Participation Diffusion Model

Abstract

This short note formalizes the visual encoding used by the interactive companion to *Participation Diffusion Dynamics in Influence-Driven Networks*. It defines the map from a student’s scalar motivation state to a display color, records the perceptual rationale for the choice of color scale, distinguishes core-network nodes from public-body display dots, and defines the aggregate and ensemble readouts that summarize the motivational state of the network at a glance. The note is self-contained and introduces no changes to the underlying model.

1 Setup

Let the club consist of n students indexed by $i \in \{1, \dots, n\}$, embedded in a weighted directed influence network with in-neighborhoods $\mathcal{N}(i)$. At each discrete time (week) t , the model assigns student i a bounded *motivation* state

$$m_i(t) \in [0, 1], \quad (1)$$

together with a *loyalty* stock $L_i(t) \in [0, 1]$ and a *fatigue* stock $F_i(t) \in [0, 1]$. We treat (1) as given by the model dynamics; the present note concerns only how $m_i(t)$ and $L_i(t)$ are rendered, and how they are summarized across i .

Because $m_i(t)$ is already normalized to the unit interval, no instance-dependent rescaling is required: the encoding below is a fixed function of $m_i(t)$, so colors are comparable across students, across weeks, and across separate simulation runs.

2 The color map

Anchor stops. Let $c_0, \dots, c_K \in [0, 255]^3$ be $K+1$ anchor colors in RGB space placed at equally spaced levels $\ell_k = k/K$. We use the $K=5$ *plasma* anchors of Table 1, a perceptually ordered sequence running from dark blue through magenta and orange to bright yellow.

k	level ℓ_k	R	G	B	swatch
0	0.0	13	8	135	
1	0.2	126	3	168	
2	0.4	177	42	144	
3	0.6	225	100	98	
4	0.8	252	166	54	
5	1.0	240	249	33	

Table 1: Plasma anchor stops c_k at levels $\ell_k = k/5$.

Piecewise-linear interpolation. Define the color map $\mathcal{C} : [0, 1] \rightarrow [0, 255]^3$ by linear interpolation between adjacent anchors. For $m \in [0, 1]$, let $u = Km$, $k = \lfloor u \rfloor$ (capped at $K-1$), and $f = u - k$; then

$$\mathcal{C}(m) = (1 - f) \mathbf{c}_k + f \mathbf{c}_{k+1}. \quad (2)$$

A core-network student with motivation $m_i(t)$ is drawn as a disk filled with $\mathcal{C}(m_i(t))$. The map $\mathcal{C}$ is continuous and applied componentwise, so small changes in motivation produce small, smooth changes in color.

Perceptual rationale. The plasma sequence is approximately *perceptually uniform* and increases monotonically in luminance from $\mathbf{c}_0$ to $\mathbf{c}_5$. Two consequences matter for reading the display. First, ordering is preserved: if $m_i < m_j$ then student i appears uniformly “cooler/darker” than student j , so relative motivation can be judged by brightness alone. Second, equal differences in m correspond to roughly equal perceived color differences, so the spatial *gradient* of color across the network is an honest proxy for the spatial gradient of motivation, rather than exaggerating or compressing particular ranges. The map degrades gracefully to grayscale luminance, preserving the ordering for color-vision deficiencies.

Core-network ring and loyalty overlay. Every simulated core-network student is marked by an outer ring so that the viewer can distinguish modeled agents from the wider public-body dots. Motivation remains the fill channel: dormant or low-motivation students are dark, while motivated students move along the plasma scale. Loyalty is rendered on a separate, non-competing channel: an amber arc of angular extent $2\pi L_i(t)$ is drawn around the node. Motivation (fill hue and brightness), core membership (outer ring), and loyalty (arc sweep) are thus separable visual variables. Inactive students — those whose participation indicator has lapsed — are desaturated and dashed, keeping them visible without treating them as current participants.

3 Core-network and public-body display layers

The mathematical model is defined on the simulated core network $V = \{1, \dots, n\}$. The interactive display may also show a surrounding field of public-body dots representing students outside that core network. These dots are a presentation layer: they help communicate that the club is embedded in a larger school population, but they are not additional model agents unless explicitly included in V . For this reason public-body dots are rendered without the core-network ring. If a public-body dot is shown as “drawn in,” that status is a qualitative display readout derived from turnout, events, and acquaintance reach; it should not be confused with the state variables m_i, L_i, F_i governed by the model equations.

The display therefore uses a simple visual grammar:

- filled disk color $\mathcal{C}(m_i)$: current motivation of a simulated core-network agent;
- outer ring: membership in the simulated core network;
- amber arc: loyalty stock L_i ;
- dashed/desaturated node: lapsed or inactive core-network agent;
- plain faint dot: surrounding public-body display dot.

4 Aggregate reach scalars

To summarize the network’s motivational state at week t in a single number we use the mean motivation, which we call the *reach* (or affectedness) of the diffusion process:

$$\bar{m}(t) = \frac{1}{n} \sum_{i=1}^n m_i(t) \in [0, 1]. \quad (3)$$

Equation (3) is the spatial average of the field that $\mathcal{C}$ colors; it rises as motivation spreads through $\mathcal{N}(\cdot)$ and falls as it drains, and it is directly comparable across runs because each $m_i \in [0, 1]$.

It is sometimes useful to count only students whose motivation has crossed a salience threshold $\tau \in [0, 1]$. Define the *active reach fraction*

$$\rho_\tau(t) = \frac{1}{n} |\{i : m_i(t) > \tau\}| \in [0, 1], \quad (4)$$

the share of the network rendered at or above the color level $\mathcal{C}(\tau)$. The companion uses a fixed τ to report a “reach” percentage; (4) is monotone nonincreasing in τ , with $\rho_0(t)$ counting all students with any motivation and $\rho_\tau(t) \leq \bar{m}(t)/\tau$ by Markov’s inequality.

A complementary scalar is the cross-sectional spread of motivation,

$$s(t) = \sqrt{\frac{1}{n} \sum_{i=1}^n (m_i(t) - \bar{m}(t))^2}, \quad (5)$$

which is small when the network is uniform (all nodes a single color — either a dim, collapsing club or a fully saturated one) and largest during the transient phases when a motivated core coexists with an unconvinced periphery. Reading $\bar{m}(t)$ together with $s(t)$ distinguishes a genuinely thriving, differentiated network from one that is uniformly saturated, even though both can share a high attendance count.

5 Sample and ensemble readouts

The companion provides two numerical views of the same equations. A sample run displays one realization of the randomized graph and heterogeneous initial conditions. An ensemble run averages M independent realizations under the same intervention path. For any scalar network quantity $Z(t)$, such as attendance $A(t)$, mean motivation $\bar{m}(t)$, or active reach $\rho_\tau(t)$, the reported ensemble mean is

$$\hat{Z}_M(t) = \frac{1}{M} \sum_{r=1}^M Z^{(r)}(t), \quad t = 0, \dots, T. \quad (6)$$

The time scrubber in ensemble mode moves through these averaged weekly states. Uncertainty bands are empirical quantiles of $\{Z^{(r)}(t)\}_{r=1}^M$ and are visual summaries of random-network sensitivity, not additional dynamical variables.

When the network panel displays averaged nodes, the corresponding value for a core-network index is

$$\hat{m}_i(t) = \frac{1}{M} \sum_{r=1}^M m_i^{(r)}(t), \quad (7)$$

and analogously for L_i, F_i , and attendance probability. Such node-level values should be read as averaged positions in the simulated core network, not as predictions about named students. This preserves the original model while allowing the interface to show both a mechanism-level sample and a robustness-oriented ensemble summary.

6 Summary

The display encodes the model’s per-student motivation $m_i(t)$ through the fixed, perceptually ordered map $\mathcal{C}$ of (2), with core-network membership marked by an outer ring and loyalty $L_i(t)$ on an orthogonal arc channel. The scalars $\bar{m}(t)$, $\rho_\tau(t)$, and $s(t)$ of (3)–(5) summarize, respectively, how far motivation has spread, what fraction of the network it has reached, and how unevenly it is distributed. The ensemble quantities (6)–(7) average these same model outputs over randomized graph realizations; they do not add or modify behavioral equations.